

Hardware Domain Solution Report

Technical Council Inductions 2026

Submission Links

Required Link	Status
Wokwi simulation link	https://wokwi.com/projects/468694734454226945
Google Drive mechanics folder link	https://drive.google.com/drive/folders/1sCTAR8JIW_sERKaiwveZCUEvJWEQhshM?usp=sharing

The local solution files are complete and organized. The links above point to the working Wokwi electronics simulation and the shared Google Drive folder for the mechanics/CAD files.

Task 1 - Ideation

Initiative Name

NIT-T Hardware Foundry

One-Line Pitch

NIT-T Hardware Foundry is a 300-student beginner-to-builder prototype accelerator where first-years turn curiosity into working campus hardware solutions, then grow into the next generation of Technical Council builders and mentors.

Problem

Many first-year students are curious about electronics, robotics, embedded systems, CAD, mechanisms, and product design, but hardware often feels intimidating because mistakes are physical. A beginner may face loose wiring, sensor noise, unclear power connections, bad tolerances, tool safety, or the fear of damaging components.

This prevents many capable students from even trying hardware early.

The Foundry addresses three barriers:

- They do not know where to begin.
- They are afraid of failing or breaking things.
- They do not have a safe, exciting, guided path from curiosity to a working prototype.

Solution

Hardware Foundry combines live demos, reusable hardware kits, studio-cluster build sessions, mentor clinics, campus-problem project tracks, failure logs, documentation, and a public prototype expo.

The goal is not to make every student build the same circuit. The goal is to give students a safe technical boundary and then let their own ideas flow. Teams choose problems, propose solutions, simplify them into minimum working prototypes, and receive support whenever they get stuck.

The full end-to-end plan is included separately as:

- NITT_Hardware_Foundry_Ultimate_Guide.pdf

- NITT_Hardware_Foundry_Ultimate_Guide.md

Target Audience

- 300 first-year students in Year 1.
- Students from all branches, including absolute beginners.
- Students interested in electronics, robotics, embedded systems, CAD, sustainability, accessibility, product design, or campus problem-solving.
- Students who may become future Technical Council hardware contributors and mentors.

Scale and Three-Year Goal

Year	Student Reach	Teams	Goal
Year 1	300	60	Prove the studio-cluster model and create the first Foundry Fellows
Year 2	500	100	Expand through trained Fellows, better kits, and more tracks
Year 3	700	140	Make Hardware Foundry a campus-wide beginner hardware pipeline

Operating Model

The event uses a distributed studio system instead of trying to place all 300 students into one build room.

- 300 students.
- 60 teams.
- 5 students per team.
- 10 studio clusters.
- 6 teams per studio cluster.
- All teams attend the launch and expo.
- Hands-on build sessions happen in staggered studio clusters.

Core Organizing Team

The initiative can be conducted by fewer than 20 core organizers.

Role	Count	Responsibility
Program Director	1	Overall ownership, permissions, final quality
Operations Lead	1	Calendar, venue schedule, participant flow
Electronics Stream Leads	3	Sensors, Arduino/ESP32, displays, debugging
Mechanics/CAD Stream Leads	3	CAD, mechanisms, mounting, enclosures
Systems Integration Leads	2	Power, wiring, reliability, test plans
Safety and Inventory Leads	2	Kits, component checkout, safety
Studio Managers	3	Coordinate studio clusters and clinics
Documentation and Design Leads	2	Project cards, posters, reports, demo storytelling
Expo and Outreach Lead	1	Publicity, final showcase, guest coordination
Total	18	Under the 20-person task limit

Venue Plan

All venues are permission-dependent candidates and must be confirmed officially before execution.

Venue Type	Candidate	Use
Launch venue	Orion / Lecture Hall Complex	Opening, demo night, safety briefing
CAD/simulation venue	Octagon / computer facilities	CAD, Wokwi, documentation, BOM clinics
Fabrication exposure	Central Workshop	Supervised fabrication demo and prototype guidance
Build rooms	Department classrooms / seminar rooms	Studio-cluster build sessions
Public expo	SAC / student activity space	Final prototype showcase
Continuation support	CEDI / incubation ecosystem	Optional support for promising prototypes

Timeline

Phase	Work
Preparation, 3 weeks	Permissions, venues, publicity, registrations, kits, project handbook
Week 1	Opening, safety, live demos, first circuit and sensor sessions
Week 2	Sensors, actuators, CAD, system thinking, debugging basics
Week 3	Idea flow week: campus problem walk, team proposal, mentor review
Week 4	Build sprint: electronics, CAD, systems, and documentation clinics
Week 5	Reliability, enclosure/mounting, test logs, demo rehearsal
Week 6	Foundry Expo, awards, archive, Fellow selection

Budget Estimate

Tier	Estimated Cost	Use Case
Lean	INR 60,000	Basic reusable kits and minimal fabrication
Recommended	INR 1,20,000	Strong 300-student execution with 60 reusable team kits
Premium	INR 2,00,000	Sponsorship-backed version with better tools, displays, enclosures, and awards

The cost is feasible because the kits are reusable and become the inventory base for Year 2 and Year 3.

Project Tracks

- Hostel Intelligence.
- Lab Helper Devices.
- Campus Sustainability.
- Accessibility and Inclusion.
- Micro-Automation.
- Open Wildcard for strong student-originated ideas.

Final Prototype Family

The broad project family is called Campus Utility Sentinel.

Each team builds a small hardware system that:

- Senses one physical condition.
- Makes one decision.
- Communicates one useful output.
- Has a safe enclosure or mounting plan.
- Includes a bill of materials.
- Includes a failure log and next-version plan.

Expected Outcomes

- 300 first-year students experience hands-on hardware.
- 60 project teams present working or partially working prototypes.
- 60 project folders are archived for future cohorts.
- 30 students are selected as Foundry Fellows for the next year.
- 10 prototypes are shortlisted for continuation.
- The initiative grows toward 700 students per year by Year 3.

Why It Fits Technical Council

Hardware Foundry directly supports Technical Council's goal of making technology accessible. It is beginner-friendly, economically feasible through reusable kits, scalable through a Fellow pipeline, and capable of creating future hardware leaders rather than one-time workshop attendees.

Task 2 - Electronics

Project Name

LCD Campus Runner

Files

File	Purpose
Electronics_Infinite_Runner/sketch.ino	Arduino game code
Electronics_Infinite_Runner/diagram.json	Wokwi circuit
Electronics_Infinite_Runner/wokwi.toml	Wokwi project config
Electronics_Infinite_Runner/README.md	Import steps, pin map, notes

What It Does

LCD Campus Runner is a two-button infinite runner game on a non-I2C 16x2 LCD. The player is shown as #, obstacles are shown as *, and the player changes row using the lane button. The start/restart button starts the game, restarts after game over, and is the only wake source after power-down sleep.

Controls

Control	Arduino Pin	Function
Lane button	2	Toggle player between upper and lower LCD row
Start/restart button	3	Start game, restart game, wake MCU from sleep

LCD Pin Map

LCD Pin	Arduino Pin
RS	12
E	11
D4	5
D5	4
D6	7
D7	8

Requirement Mapping

Requirement	Status	Notes
Level 0: LCD wiring	Complete	Non-I2C 4-bit LCD wiring is used. Serial Monitor text is echoed to the LCD before the game starts.
Level 1: controllable object	Partially adapted	Final task needs only row switching, so the final build uses one lane button instead of four direction buttons. Input is interrupt-based and non-blocking.
Level 2: obstacle runner	Complete	Obstacles move right to left, collision detection is implemented, and the game stops on collision.
Level 2: no unavoidable two-row wall	Complete	Only one obstacle is active at a time, so the player always has a possible safe row.
Level 2/3: power-down mode	Complete	After game over, the MCU enters power-down sleep and the lane interrupt is disabled, so only the restart button wakes it.
Level 3: game-over screen	Complete	LCD shows Game Over! plus score and high score.
Level 3: high score retained while powered	Complete	High score is stored in RAM and preserved while the MCU remains powered.
Level 3: true random number attempt	Implemented with limitation	The seed combines floating analog-pin LSB samples with timer jitter. A real circuit can improve this with a physical noise source.
Brownie: non-I2C LCD	Complete	Uses LiquidCrystal with parallel pins, not an I2C backpack.
Brownie: non-blocking buttons	Complete	Buttons use external interrupts; frame timing uses millis().
Brownie: manual LCD control without library	Not attempted	I chose LiquidCrystal to keep the final solution simple and reliable.

Challenges and Solutions

Challenge	Solution
Avoiding blocking button reads	Used interrupts on pins 2 and 3.
Preventing impossible obstacle patterns	Generated only one obstacle at a time.
Waking only by restart button	Detached the lane interrupt before entering power-down sleep and reattached it after wake.
Randomness in simulation	Combined analog noise sampling with timer jitter and documented the Wokwi limitation.

Wokwi Instructions

1. Open Wokwi and create an Arduino Uno project.
2. Replace the generated sketch.ino with Electronics_Infinite_Runner/sketch.ino.

3. Replace diagram.json with Electronics_Infinite_Runner/diagram.json.
4. Run the simulation.
5. Paste the Wokwi share link in the Submission Links table at the top of this report before uploading.

Task 3 - Mechanics

Project Name

Parallel Dual Scissor Lift

Files

File	Purpose
Mechanics_Scissor_Lift/scissor_mechanism.scad	Parametric OpenSCAD CAD model
Mechanics_Scissor_Lift/scissor_motion_study.gif	Local rendered motion study animation
Mechanics_Scissor_Lift/scissor_motion_study_preview.png	Preview image
Mechanics_Scissor_Lift/generate_motion_study.py	Script used to generate the GIF
Mechanics_Scissor_Lift/README.md	CAD usage and upload instructions

Requirement Mapping

Requirement	Status	Notes
Level 1: single scissor link	Complete	Link is 150 mm x 30 mm with 10 mm end and center pivot holes.
Level 1: pins	Complete	Pin module is included with 9.8 mm practical clearance for 10 mm holes.
Level 2: rectangular base plate	Complete	Base includes fixed pivot holes and linear slots for sliding pivots.
Level 2: top platform	Complete	Top platform includes corresponding pivot holes.
Level 2: dual parallel X-units	Complete	CAD assembly includes two X-units and bracing rods.
Level 3: motion study	Complete locally	A rendered GIF demonstrates extension and retraction.
Google Drive upload	Complete	The shared Drive folder link is included in the Submission Links table above.

CAD Design Notes

- Link dimensions: 150 mm x 30 mm x 6 mm.
- Pivot hole diameter: 10 mm.
- Pin diameter: 9.8 mm for assembly clearance.
- Base plate: rectangular plate with two fixed pivot locations and two slotted guide paths.
- Top platform: rectangular plate with upper pivot mounting points.
- Cross bracing: rods connect the parallel units to keep the mechanism synchronized.

Motion Study

The included GIF shows a simple kinematic extension and retraction sequence. For a stricter CAD-software motion study, the same geometry can be recreated or imported into Fusion 360/SolidWorks, using revolute joints at circular pivots and slider joints at the slotted pivots.

Google Drive Instructions

The mechanics folder has been uploaded and shared through the link in the Submission Links table. The local Mechanics_Scissor_Lift/ folder remains included with this packet so the evaluator can also inspect the editable CAD/source files directly.

Final Checklist

Item	Status
Ideation document content	Complete
Electronics Wokwi-ready circuit and code	Complete
Electronics implementation report	Complete
Mechanics CAD source	Complete
Mechanics motion study GIF	Complete
Wokwi link pasted into report	Complete
Google Drive link pasted into report	Complete